

Milestone 6 Documentation

Milestone 6 focused on implementing the power injection and mismatch calculations needed before using the Newton-Raphson method. In this milestone, the Settings class was expanded so it could do more than just store system values. It now includes methods for computing bus power injections, building the mismatch vector, and generating a mismatch table. This made the class a central place for handling the main calculations needed for power flow setup.

Main implementation parts

- Store system frequency and base apparent power in the Settings class
- Compute real and reactive power injection at each bus
- Compute the mismatch vector for all non-slack buses
- Handle slack, PV, and PQ buses differently
- Build a table showing bus mismatch values in MW, Mvar, and MVA

Power injection implementation

The compute_power_injection() method calculates the real and reactive power at a bus using the voltage magnitudes, bus angles, and the Ybus matrix. Inside the method, the code loops through every bus in the system and uses the real and imaginary parts of the admittance matrix to calculate P_i and Q_i . The bus angles are converted from degrees to radians before using the trig functions. This method returns the calculated real and reactive power for the selected bus.

Complex power equation:

$$S_i = V_i \sum_{j=1}^n Y_{ij}^* Y_j^*$$

Real power equation:

$$P_i = |V_i| \sum_{j=1}^n |V_j| (G_{ij} \cos(\delta_i - \delta_j) + \beta_{ij} \sin(\delta_i - \delta_j))$$

Reactive power equation:

$$Q_i = |V_i| \sum_{j=1}^n |V_j| (G_{ij} \sin(\delta_i - \delta_j) - \beta_{ij} \cos(\delta_i - \delta_j))$$

Mismatch calculation

The `compute_power_mismatch()` method builds the mismatch vector by comparing specified power values to calculated power values. For each bus except the slack bus, the code finds the calculated injection values first, then determines the specified values by checking connected generators and loads. Generator real power is added, while load real and reactive powers are subtracted. After that, the mismatch is found by subtracting calculated power from specified power.

Real power mismatch:

$$\Delta P_i = P_i^{spec} - P_i^{calc}$$

Reactive power mismatch:

$$\Delta Q_i = Q_i^{spec} - Q_i^{calc}$$

For bus handling:

- **Slack bus**
 - Skipped in the mismatch vector
- **PV bus**
 - Includes only real power mismatch
- **PQ bus**
 - Includes both real and reactive power mismatch

Mismatch table

The `compute_mismatch_table()` method was also added to make the results easier to read. This method calculates the mismatch at every bus and puts the results into a pandas DataFrame. The table includes the bus number, name, type, and mismatch values in MW, Mvar, and MVA. For PV buses, reactive mismatch is set to zero, and for the slack bus both mismatches are set to zero. The apparent mismatch is found using the magnitude of the real and reactive mismatch values.